

MUHAMMAD SIBYAN NAFIL ILMI

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Fresh Graduate in Mechatronics Engineering Technology from Politeknik Negeri Jember (GPA 3.93 / 4.00) with hands-on experience in Industrial Automation, Embedded Systems, Robotics, PLC and HMI Programming, SCADA Integration, and IoT (Internet of Things). Gained practical experience through applied projects and research, including IoT-based monitoring systems, smart farming technologies, and conveyor systems. Experienced in end-to-end system development, covering system design, microcontroller/PLC and HMI programming, as well as system testing and performance evaluation. Skilled in integrating mechanical, electrical, and software components to develop functional and efficient systems.

Education Level

Politeknik Negeri Jember - Jember, East Java	Sep 2022 - Jun 2026
D4 Mechatronics Engineering Technology GPA: 3.93 / 4.00	
- Thesis: Development of a Path Planning Algorithm for a Raspberry Pi-Based 4-Wheeled Omnidirectional Cellular Conveyor System - Relevant Courses: IoT, Robotics, PLC, SCADA & HMI, Sensor & Transducer, Pneumatic, CAD & CAM, microcontroller, motor and actuator control system	

Work Experience

PT. Global Andalan Solusi Sukses – Surabaya, East Java	
Production Staff Intern	Aug 2025 – Nov 2025
Responsibilities & Contribution:	
- Manufactured and assembled various speed limiter devices for vehicle safety systems. - Diagnosed and resolved manufacturing defects through root-cause analysis, minimizing downtime and maintaining production continuity - Performed quality control inspections on sensors, actuators, and mechanical assemblies to ensure compliance with production standards and prevent defective units from reaching customers - Installed speed limiter systems on-site, including wiring harness integration and cable mapping for vehicle electrical systems	

Project Experience & Achievements

Final Years Thesis Project	Jan 2026 – May 2026
Title: Development of a Path Planning Algorithm for a Raspberry Pi-Based 4-Wheeled Omnidirectional Cellular Conveyor System	
Responsibilities & Contribution:	
- Architected a distributed embedded control system using Raspberry Pi 5 as the I2C master and multiple ESP32 microcontrollers as slave node, each controlling individual conveyor cells. - Calibrated drive motor RPM across all conveyor cells, achieving up to 97% accuracy and improving path-tracking consistency. - Designed and implemented Dijkstra’s algorithm for path planning on an 8-directional weighted grid graph. - Developed a real-time GUI monitoring interface for system visualization and control. - Implementing ArUco marker-based visual tracking using OpenCV to track the movement of a box.	
Student Creativity Program – Innovative Creation 2024 (PKM-KC 2024)	Feb 2024 – Oct 2024
Title: Soil Nutrient Monitoring: Design and Development of an Information Technology-Based Soil Quality Detector as an Innovation in the Agricultural Sector	
Achievement: Grant recipient, PKM-KC 2024 — Finalist, 37th Pekan Ilmiah Mahasiswa Nasional (PIMNAS-37).	
Responsibilities & Contribution:	
- Led hardware development from start to finish, encompassing system architecture design based on problem requirements, 3D hardware design, PCB schematic design, microcontroller programming, hardware fabrication and assembly, and sensor calibration. - Calibrated multiple sensors (NPK, pH, temperature, and soil moisture), achieving accuracy levels above 95% - Conducted functional testing and validation to ensure the hardware system operated reliably and met intended performance requirements	
5th Agricultural Innovation and Technology Competition (AITeC 5)	Jul 2023 – Oct 2023
Title: Automatic Nutrition Mixing Innovation for IoT-Based Hydroponic Plant Cultivation	
Achievement: 2nd Place, Effective Category — Agricultural Technology Innovation Competition	
Responsibilities & Contribution:	
- Selected, installed, calibrated, and tested TDS, water-level, and water-flow sensors. - Designed electrical circuits, determined wiring configurations, power supply requirements, actuator drivers, and component integration. - Conducted system testing, sensor calibration, nutrient dosing validation, and overall system performance evaluation.	

Skills & Certification

Tools

- CAD : Autodesk Inventor, Autocad
- PCB Design : Kicad, Eagle, Altium Designer
- PLC : CX-Programmer, EcoStruxure Machine Expert
- HMI & SCADA : Vijeo Designer, Haiwell Cloud Scada
- Embedded & Firmware Development : Arduino IDE, PlatformIO, STM32CubeIDE

Technical Skills

- PLC Programming, HMI Configuration & SCADA Integration
- Instrumentation & Control Systems
- Embedded Systems Development
- IoT System Development
- Panel Assembly and Electrical Installation
- Pneumatic Systems

Soft Skills

- Leadership & Team Coordination
- Project Planning & Management
- Cross-functional Collaboration
- Problem Solving & Analytical Thinking

Certification

- Certificate (2026 - 2029): Mechatronics Engineer Certification – Badan Nasional Sertifikasi Profesi (BNSP)